

House Price Prediction – Simple Summary

This project is about predicting house prices using machine learning. The dataset contains 545 house records with 13 features. The target variable is price. The data includes both numerical and categorical information such as area, bedrooms, bathrooms, parking, air conditioning, and other house features.

First, the data was explored to understand its structure and quality. The analysis showed that there were no missing values and no duplicate records in the dataset. Categorical variables were converted into numerical values using One-Hot Encoding so that machine learning models could process them.

Exploratory Data Analysis (EDA) was performed to understand the relationship between different features and house prices. The histogram showed that house prices are slightly right-skewed, meaning most houses have low to medium prices, while only a few houses are very expensive. The correlation heatmap showed that area, bathrooms, air conditioning, stories, and parking have a strong positive relationship with house prices. The scatter plot also showed that larger houses generally have higher prices.

The dataset was divided into training data (80%) and testing data (20%). Two machine learning models were trained and compared: Linear Regression and Random Forest Regressor. Their performance was evaluated using MAE, RMSE, and R^2 score.

The results showed that Linear Regression performed better than Random Forest. It produced lower prediction errors and achieved a higher R^2 score of about 0.65. Therefore, Linear Regression was selected as the final model for house price prediction.

Overall, the project found that house size, number of bathrooms, parking facilities, air conditioning, and number of stories are the most important factors affecting house prices. These findings can help real estate companies and buyers understand what features increase the value of a property.